PCA (Class Notes)

Saturday, 27 June 2026 1:46 pm

- Principal Component Analysis, or PCA, is a technique that reduces complex data into fewer dimensions while keeping as much important variation as possible.
- It's like summarizing a whole book into a few powerful sentences without losing the meaning.

Problem: FACE RECOGNITION

↳ Facial feature analysis using a 3D geometric signature instead of raw pixels.

Feature 1: (x_1) → Eye gap (distance b/w inner corners of eyes in mm).

Feature 2: (x_2) → Nose length (vertical length of nose bridge in mm)

Feature 3: (x_3) → Mouth width (horizontal width of lips in mm).

Sample size = N = 3 face images

Target Goal: reduce face representation from 3D to 1D using PCA.

	E	N	L
F1	–	–	–
F2	–	–	–
F3	–	–	–

Raw Dataset & Mean Centering

Raw input data vectors:

$$x_1 = \begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix} \begin{matrix} \to \text{eye gap of face 1} \\ \to \text{nose length of face 1} \\ \to \text{mouth width of face 1} \end{matrix}$$

Similarly for face 2 & face 3:

$$x_2 = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix} \qquad x_3 = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}$$

$$\begin{array}{c|c} & \begin{matrix} F_1 & F_2 & F_3 \end{matrix} \\ \begin{matrix} E \\ N \\ M \end{matrix} & \begin{pmatrix} 2 & 4 & 3 \\ 2 & 5 & 2 \\ 5 & 3 & 4 \end{pmatrix} \end{array}$$

$x_1 + x_2 + x_3$

Sample Mean vector (μ): $\mu = \frac{1}{N} \sum_{i=1}^{N} x_i$

$$\mu = \frac{1}{3} \sum_{i=1}^{3} x_i$$

$$\mu = \frac{1}{3}\left(\begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix} + \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}\right) = \frac{1}{3}\begin{bmatrix} 9 \\ 9 \\ 12 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix} \begin{matrix} \to E \\ - N \\ - L \end{matrix}$$

Average Face of this dataset.

Mean = $x - \bar{x}$

① Mean vector

② Mean center deviation

↓

Central data matrix (X)

③ Co variance mat

$$S = \frac{1}{N-1} X \cdot X^T$$

④ $|S - \lambda I| = 0$

λ: ...

Mean $= \check{x} - \bar{x}$

(4) $|S - \lambda I| = 0$

$\lambda_i \ldots$

(5) Eigen vector

Mean-Centered Deviations: $(x_i' = x_i - \mu)$

$$x_1' = \begin{bmatrix} 2-3 \\ 2-3 \\ 5-4 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

$$x_2' = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \quad \& \quad x_3' = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

→ how each individual face deviates from the baseline norm (shifting the dataset coordinate system center directly to the origin.)

3 feature, 3 F N

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ \end{array}$$
$$\begin{bmatrix} -1 & -2 & 6 \\ 2 & -1 & 2 \\ 1 & 3 & 2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 3 \\ 3 \\ 6 \end{bmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

-2
1
-1

Centered Data Matrix (X):

$$X = [x_1' \;\; x_2' \;\; x_3'] = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

$$\begin{array}{cccc} & x & y & z \\ x & r_{xx} & r_{xy} & r_{xz} \\ y & r_{yx} & r_{yy} & r_{yz} \\ z & r_{zx} & r_{zy} & r_{zz} \end{array}$$

Covariance Matrix (S)

Covariance Formula $= S = \frac{1}{N-1} \cdot X X^T$

$X = 3 \times 4$

$X^T = 4 \times 3$

$X \; X^T = 3 \times 3$

$5 \times T$

7×5

$\boxed{5 \times 5}$

$$S = \frac{1}{3-1} XX^T = \frac{1}{2} XX^T$$

$$XX^T = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix}\begin{bmatrix} -1 & -1 & 1 \\ 1 & 2 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$XX^T = \begin{bmatrix} 2 & 3 & -2 \\ 3 & 6 & -3 \\ -2 & -3 & 2 \end{bmatrix}$$

N L

EN = +1.5
EL = -1

$$S = \begin{bmatrix} 1 & 1.5 & -1 \\ 1.5 & 3 & -1.5 \\ -1 & -1.5 & 1 \end{bmatrix}$$

main diagonal → variances of eye gap, nose length, mouth width respectively (1, 3, 1)

off diagonal elements → covariances among 3 features.

$de(A - \lambda I)$

Eigenvalue Calculation

de (A − λI)

Eigenvalue Calculation & Component Ranking

Eigenvalues

$\det(S - \lambda I) = 0$

$$\begin{vmatrix} 1-\lambda & 1.5 & -1 \\ 1.5 & 3-\lambda & -1.5 \\ -1 & -1.5 & 1-\lambda \end{vmatrix} = 0$$

$\Rightarrow \lambda_1 = 4.68, \ \lambda_2 = 0.32, \ \lambda_3 = 0$

Take largest Eigenval

Variance Percentage Analysis

$\lambda_1 + \lambda_2 + \lambda_3 = 4.68 + 0.32 + 0 = 5$

Variance captured by $\lambda_1 = \frac{4.68}{5} \times 100 = 93.6\%$

For $\lambda_2 \rightarrow 6.4\%$ $\quad \frac{0.32}{5} \times 100$

for $\lambda_3 \rightarrow 0\%$

λ_1 contains nearly all of the mathematical information about variations in our database faces.

We can drop λ_2 & λ_3 with minimal data loss.

$\lambda_1 = 60\%$

$\lambda_2 = 40\%$

Eigen Vector (The Primary Eigenface)

$(S - 4.68I)\vec{v_1} = \vec{0}$

$$\begin{bmatrix} -3.68 & 1.5 & -1 \\ 1.5 & -1.68 & -1.5 \\ -1 & -1.5 & -3.68 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

RREF

Row Reduced Echlon form

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1.78 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \begin{array}{l} v_1 + v_3 = 0 \\ v_2 + 1.78v_3 = 0 \end{array} \Rightarrow \begin{array}{l} v_3 = t \\ v_1 = -t \\ v_2 = -1.78t \end{array}$$

$\Rightarrow \vec{v_1} = \begin{bmatrix} -1 \\ -1.78 \\ +1 \end{bmatrix} \rightarrow$ Unnormalized vector.

$$\vec{V} = \begin{pmatrix} -1 \\ -1.78 \\ 1 \end{pmatrix}$$

Vector normalization (Unit vector Scaling)

$\|\vec{v_1}\| = \sqrt{(-1)^2 + (-1.78)^2 + (1)^2} = 2.27$

normal

$\bar{V} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$

$|V| \ \sqrt{v_1^2 + v_2^2 + v_3^2}$

$$\vec{V}_{1\,normalized} = \frac{1}{2.27} \begin{bmatrix} -1 \\ -1.78 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.44 \\ -0.78 \\ 0.44 \end{bmatrix}$$

Eyegap & nose length scale downward

mouth width increases.

these measure direction only. (scale of vector is removed.)

Vector normalization (Unit vector Scaling)

$\|\vec{v_1}\| = \sqrt{(-1)^2+(-1.78)^2+(1)^2} = 2.27$ ✓

$\vec{V_1}_{normalized} = \frac{1}{\boxed{2.27}}\begin{bmatrix}-1\\-1.78\\1\end{bmatrix} = \begin{bmatrix}-0.44\\-0.78\\0.44\end{bmatrix}$

Eyegap & nose length scale downward

mouth width increases.

these measure direction only.
(Size/Scale of vector is removed.)

normal

$\bar{V} = \begin{pmatrix}v_1\\v_2\\v_3\end{pmatrix}$

$|V| \quad \sqrt{v_1^2+v_2^2+v_3^2}$

$P = V_N^T$

Dimensionality Reduction

Projection Matrix

$P = [\vec{V_1}^T_{normalized}] = [-0.44 \quad -0.78 \quad 0.44]$

1D

$y_1 = P\,x_1'$

y_1

$1\times 3 \quad 3\times 1$

$y_i = P\,x_i'$

Projection of Face 1 into 1D eigenspace

$y_1 = Px_1'$

$y_1 = [-0.44 \quad -0.78 \quad 0.44]\begin{bmatrix}-1\\-1\\1\end{bmatrix}$ mean centered vector of face 1

$y_1 = 1.66$

3D vector of face 1 is compressed into a single scalar score of 1.66

face 2

$y_2 = [-0.44 \quad -0.78 \quad 0.44]\begin{bmatrix}1\\2\\-1\end{bmatrix} = -2.44$

features deviate from average face in the opposite direction of face 1.

face 3

$y_3 = 0.78$

CS CamScanner

Interpretation

If $y_i = 0 \to$ face is average. (it matches the average face we calculated)

If $y_i > 0 \to$ face shifts with the direction of eigenface vector.

If $y_i < 0 \to$ face shifts in the exact opposite direction of eigenface vector.

Primary Eigenface Direction :

$$\text{Rule Vector} = \begin{bmatrix} -0.44 \\ -0.78 \\ \boxed{0.44} \end{bmatrix} \begin{matrix} \to \text{Eye Gap Rule} \\ \to \text{Nose Length Rule} \\ \to \text{Mouth width Rule} \end{matrix}$$

Face1

$y_1 = 1.66 \to$ follows the rule vector directly

- wider mouth than average
- narrower eyegap than avg.
- shorter nose than avg.

Face2

$y_2 = -2.44 \to$ flips the rule vector entirely

- narrower mouth than avg.
- wider eye gap
- longer nose

Face3

$y_3 = 0.78 \to$ follows the rule vector slightly.

0.78 is closer to 0

Face 3 is much closer to avg. face.

① Average vector $\bar{x} = \frac{1}{N}\left[\Sigma x_i\right]$

② Mean deviation $x_i - \bar{x}$

③ $\downarrow$ matrix (deviation matrix) $= \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{pmatrix}$

(2) Mean deviation $x_i - \bar{x}$

(3) $\downarrow$ matrix (deviation matrix) $= \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$

X

(4) Covariance matrix $= S$ $\quad \frac{1}{N-1} X \cdot X^T$

if $-\lambda \rightarrow +\lambda$

(5) Eigen values $\quad |S - \lambda I| = 0$

$\lambda_1, \lambda_2, \lambda_3$

Suitable λ $= \frac{\lambda_1}{\lambda_1+\lambda_2+\lambda_3}$ $\quad$ ("Highest value of λ")

for particular λ

(6) Eigen vector $\quad S - \lambda I = 0$ $\quad \begin{pmatrix} a-\lambda & 1 & 2 & 0 \\ 4 & b-\lambda & 1 & 0 \\ 2 & 1 & c-\lambda & 0 \end{pmatrix}$

$\vec{V} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$

(7) Normalize $\quad |V| = \sqrt{v_1^2 + v_2^2 + v_3^2} = K$

$V_N = \frac{1}{K}\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} v_1/K \\ v_2/K \\ v_3/K \end{pmatrix} = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}$

(8) dimension Reduct

Projection vector $\quad P = V_N^T = (w_1 \quad w_2 \quad w_3)$

$y_i = P \cdot x_i'$ $\qquad x_i' =$ mean deviation vector

$y_i = (w_1 \quad w_2 \quad w_3)\begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix}$ $\qquad y_1 = N$, $y_2 = -N$, $y_3 = -K$

$y_i = 0$ $\quad$ Average face

$y_i > 0$; deviation is in the line direction eigen vector

$y_i < 0$ $\quad$ " " opposite to eigen vector

Let Suppose $\quad \vec{V} = \begin{pmatrix} -0.231 \\ 0.521 \\ -0.718 \end{pmatrix} \begin{matrix} \rightarrow f_1 \\ \rightarrow f_2 \\ \rightarrow f_3 \end{matrix}$

f_1, f_3 increase

Let Suppose

$V = \begin{pmatrix} 0.521 \\ -0.718 \end{pmatrix} \begin{matrix} \to f_2 \\ \to f_3 \end{matrix}$

$y_1 = 2.34$ → f_1, f_3 – reduce
→ f_2 → increase

$y_2 = -3.45$ → f_1, f_3 Increase
→ f_2 decreas

A researcher is studying the physical characteristics of 4 different species of birds. They measure 3 features:

* features

Bird	Wingspan (cm)	Beak Length (cm)	Tail Length (cm)
A	25	8	15
B	35	12	20
C	30	10	19

$x_1 = \begin{pmatrix} 25 \\ 8 \\ 15 \end{pmatrix}$ $x_2 = \begin{pmatrix} 35 \\ 12 \\ 20 \end{pmatrix}$ $x_3 = \begin{pmatrix} 30 \\ 10 \\ 19 \end{pmatrix}$

① Mean vector $= \frac{1}{3}\begin{pmatrix} 90 \\ 30 \\ 54 \end{pmatrix} = \begin{pmatrix} 30 \\ 10 \\ 18 \end{pmatrix}$

② Mean deviation $= \quad x_1' = \begin{pmatrix} 25 \\ 8 \\ 15 \end{pmatrix} - \begin{pmatrix} 30 \\ 10 \\ 18 \end{pmatrix} = \begin{pmatrix} -5 \\ -2 \\ -3 \end{pmatrix}$

$x_2' = \begin{pmatrix} 35 \\ 12 \\ 20 \end{pmatrix} - \begin{pmatrix} 30 \\ 10 \\ 18 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix}$

$x_3' \quad \begin{pmatrix} 30 \\ 10 \\ 19 \end{pmatrix} - \begin{pmatrix} 30 \\ 10 \\ 18 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$X = \begin{pmatrix} -5 & 5 & 0 \\ -2 & 2 & 0 \\ -3 & 2 & 1 \end{pmatrix}$

$\Bigg(\ 2\times2 \quad X = \begin{pmatrix} 2 & 1 \\ -1 & 4 \\ 5 & 3 \end{pmatrix}$

$3\times2 \quad 2\times3$

$S = \frac{1}{2}$

$\frac{1}{N-1}$

$XX^T = \begin{pmatrix} -5 & 5 & 0 \\ -2 & 2 & 0 \\ -3 & 2 & 1 \end{pmatrix} \begin{pmatrix} -5 & -2 & -3 \\ 5 & 2 & 2 \\ 0 & 0 & 1 \end{pmatrix}$

$= \frac{1}{2} \begin{pmatrix} 50 & 20 & 25 \\ 20 & 8 & 10 \\ 25 & 10 & 14 \end{pmatrix}$

$9+4+1$

$$S = \begin{pmatrix} 25 & 10 & 12.5 \\ 10 & 4 & 5 \\ 12.5 & 5 & 7 \end{pmatrix}$$

Eigen value

S

$$|S - \lambda I| = 0 \quad \begin{vmatrix} 25-\lambda & 10 & 12.5 \\ 10 & 4-\lambda & 5 \\ 12.5 & 5 & 7-\lambda \end{vmatrix} = 0$$

$$(25-\lambda)\begin{vmatrix} 4-\lambda & 5 \\ 5 & 7-\lambda \end{vmatrix} - 10\begin{vmatrix} 10 & 5 \\ 12.5 & 7-\lambda \end{vmatrix} + 12.5\begin{vmatrix} 10 & 4-\lambda \\ 12.5 & 5 \end{vmatrix} = 0$$

$$(25-\lambda)\left((4-\lambda)(7-\lambda) - 25\right) - 10\left(70 - 10\lambda - 62.5\right) + 12.5\left(50 - 50_{-12.5\lambda}\right) = 0$$

$$\boxed{\text{Suppose } \lambda \Rightarrow 20}$$

$$\left[\begin{array}{ccc|c} 5 & 10 & 12.5 & 0 \\ 10 & -16 & 5 & 0 \\ 12.5 & 5 & -13 & 0 \end{array}\right]$$

Gauss-met

Eigen vector $\vec{V} = t\begin{pmatrix} -2.5 \\ 1 \\ 0 \end{pmatrix}$

Normalize

$$|V| = \sqrt{2.5^2 + 1} = \sqrt{6.25 + 1} = \sqrt{7.25}$$

$$V_N = \frac{1}{|V|} \cdot \vec{V} = \begin{pmatrix} -2.5/\sqrt{7.25} \\ 1/\sqrt{7.25} \\ 0 \end{pmatrix}$$

Projecti

1×3

$$P = \left(\frac{-2.5}{\sqrt{7.25}}, \frac{1}{\sqrt{7.25}}, 0 \right)$$

$y_i = P \cdot x_1'$

$y_2 = P \times x_2'$

$y_3 = P \times x_3'$

Eigen vector =

$\begin{pmatrix} -2.5 \\ 1 \\ 0 \end{pmatrix}$ → Reduce, Increase, ,

$y_1 = +ve$ same as vector

$y_2 = -ve$ opposite as vect

(F_1 = Increa, F_2 = decrea)

Eigen vector

y_i

$y_i > 0$, same direction → $\begin{pmatrix} -1.25 \\ 0.56 \\ -2.34 \end{pmatrix}$ → F_1 = decreas, F_2 = Increas, F_3 = decrease

$y_i < 0$; opposite direction

$F_1 \Rightarrow$ Increase
f_2 : decrease
f_3 : Increase

PCA Case Study: Reducing 2D Data to 1D

X	Y
2	3
3	5
4	4
5	7
6	8

$\begin{pmatrix} 4 \\ 5.4 \end{pmatrix}$

Step 1: Centering the Data

Mean $X = \bar{x} = \frac{2+3+4+5+6}{5} = 4$

Mean $Y = \bar{y} = \frac{3+5+4+7+8}{5} = 5.4$

- **Mean X (μ_x):** $\frac{2+3+4+5+6}{5} = 4$
- **Mean Y (μ_y):** $\frac{3+5+4+7+8}{5} = 5.4$

Centered Data Table ($X_{centered} = X - \mu$):

Original (x, y)	Centered $(x-4, y-5.4)$
(2,3)	(−2.0, −2.4)
(3,5)	(−1.0, −0.4)
(4,4)	(0.0, −1.4)
(5,7)	(1.0, 1.6)
(6,8)	(2.0, 2.6)

$X - \bar{X}$	$Y - \bar{Y}$
−2	−2.4
−1	−0.4
0.0	−1.4
1	1.6
2	2.6

2×5

5×2

$Var(x) = \frac{4+1+0+1+4}{4} = \frac{10}{4} = 2.5$

$Var(y) = \frac{(-2.4)^2 + (-0.4)^2 + (1.4)^2 + 1.6^2 + 2.6^2}{4}$

$= 4.3$

X	Y
2	3
3	5
4	4
5	7
6	8

x_1	x_2
2	3
3	5
4	4
5	7
6	8

$x_1 + x_2$

$\begin{pmatrix} 4 \\ 5.4 \end{pmatrix}$

2	3
3	5
4	4
5	7
6	8

$$\begin{matrix} 2 & 3 \\ 3 & 5 \\ 4 & 4 \\ 5 & 7 \\ 6 & 8 \end{matrix} \qquad x_1 + x_2 \qquad \begin{pmatrix} 4 \\ 5.4 \end{pmatrix}$$

$x \cdot \quad X^T$

$$\begin{array}{cc} X-\bar{X}, & Y-\bar{Y} \end{array}$$

$$\begin{pmatrix} -2 & -2.4 \\ -1 & -0.4 \\ 0.0 & -1.4 \\ 1 & 1.6 \\ 2 & 2.6 \end{pmatrix} \quad 2\times5$$

$$\begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ -2.4 & -0.4 & -1.4 & 1.6 & 2.6 \end{pmatrix} \begin{pmatrix} -2 & -2.4 \\ -1 & -0.4 \\ 0 & -1.4 \\ 1 & 1.6 \\ 2 & 2.6 \end{pmatrix}$$

$$\frac{1}{4}\begin{pmatrix} \underline{4+1+0+1+4} & 12 \\ 12 & 17.2 \end{pmatrix}$$

$$= \frac{1}{4}\begin{pmatrix} 10 & 12 \\ 12 & 17.2 \end{pmatrix} = \begin{pmatrix} 2.5 & 3 \\ 3 & 4.3 \end{pmatrix}$$

$\underline{S}$

*

x_1 / X	x_2 / Y	Σx_i	$\frac{1}{5}\Sigma x_i$
-1	2	1	0.2
3	1	4	0.8
5	-2	3	0.6
4	2	6	1.2
6	7	13	2.6

$$\bar{X} = \begin{pmatrix} 0.2 \\ 0.8 \\ 0.6 \\ 1.2 \\ 2.6 \end{pmatrix}$$

$$X = \begin{pmatrix} -1.2 & 1.8 \\ 2.2 & 0.2 \\ 4.4 & -2.6 \\ 2.8 & 0.8 \\ 3.4 & 4.4 \end{pmatrix} \quad 5\times2$$

X^T 2×5

$X^T X$

$2\times5 \quad 5\times2$

$$X^T \begin{pmatrix} -1.2 & 2.2 & 4.4 & 2.8 & 3.4 \\ \cdot & & & & \end{pmatrix} \begin{pmatrix} -1.2 & 1.8 \\ 2.2 & 0.2 \\ \cdot & \cdot \end{pmatrix}$$

$$X^T X = \begin{pmatrix} -1.2 & 2.2 & 4.4 & 2.8 & 3.4 \\ 1.8 & 0.2 & -2.6 & 0.8 & 4.4 \end{pmatrix} \begin{pmatrix} 2.2 & 0.2 \\ 4.4 & -2.6 \\ 2.8 & 0.8 \\ 3.4 & 4.4 \end{pmatrix}$$

$\underline{x_1 = (2, 2, 5)}$ $\quad$ $x_2 = \underline{(4, 5, 3)}$ $\Rightarrow$ Apply PDf Examp of face on x_1 x_2 $\begin{pmatrix} 2 & 4 \\ 2 & 5 \\ 5 & 3 \end{pmatrix}$